Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

1 The decision

An indie studio's creative director chooses the next game's two tags. If no pair passed the rule in section 5, they would take a crowded pair (table, last row).

2 Why a network

Counts can't show whether two tags draw different players. The network can: tags with few shared neighbours point to audiences that rarely meet.

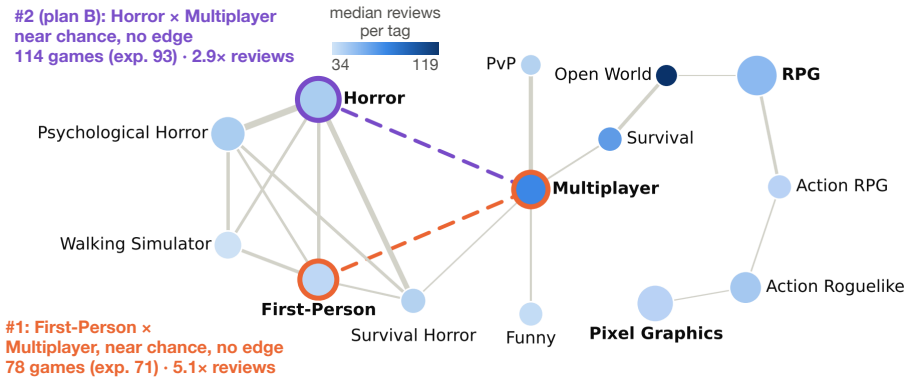
3 Data

Source	S1 FronkonGames (Hugging Face); S2 live Steam store.
Obtained	S1 download (Sep 2025). S2: 1,000 random S1 games (seed 7202), scraped 2026-10-14.
Legal	S1 MIT licence. S2: robots.txt allows /app/, /appreviews/; 1 req/1.5 s; no personal data.
Period	Jan 2023 – Dec 2024 releases.
Volume	S1: all 9,094 Indie games with ≥10 reviews. S2: 996 with a live page.
Check	S1 vs S2: mean top-5 tag Jaccard 0.87; Spearman ρ of reviews 0.98.
Bias	Favours games that sold; reviews = buyers, not unmet demand; late-2024 games had less time.

4 Graph construction

Node	A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph.
Edge	Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation.
Resolution	Co-op, Online/Local Co-Op, Local Multiplayer → Multiplayer (84% of Horror × Multiplayer games were co-op too).
Weight	NPMI: co-occurrence vs chance (0 = chance). By raw count, 9 of the top-10 edges touch the 5 commonest tags; by NPMI, 0.
Window	Top 5 tags per game (as on the store page). With all 20, #1 spans 375 games and its reviews drop from 5.1× to 3.0×.
Excluded	"Indie"; Early Access, Free to Play, adult; on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration.
Pruning	Tag on ≥1.5% of games (136); edge needs NPMI ≥ 0.10 and ≥0.5% of games (45) with both tags. 439 tags → **80 nodes, 177 edges**.

5 The picture



First-Person and Multiplayer share only Survival Horror. 14 of 80 tags: ranked pairs (bold), the 2 strongest neighbours of #1 and #2, linking paths. Size = games; width = NPMI; *not shown.

Selection rule: among pairs on ≥73 games, keep those **not crowded** (games ≤ 1.5× chance; chance = games with A × games with B ÷ 9,094) with tags in **separate neighbourhoods** (≤20% shared neighbours); rank by median reviews vs the median game (40).

Rank: pair	Games (chance)	Crowding	Shared neighbours	NPMI	Median reviews
#1: **First-Person × Multiplayer**	78 (71)	1.10×	8%	0.02	204 (5.1×)
#2 (plan B): **Horror × Multiplayer**	114 (93)	1.23×	8%	0.05	117 (2.9×)
#3: **Pixel Graphics × RPG**	133 (107)	1.25×	0%	0.05	70 (1.8×)
Crowded*: **Building × Simulation**	74 (29)	2.54×	8%	0.19	136 (3.4×)

6 Findings

- **#1 First-Person × Multiplayer** is near chance (NPMI 0.02, no edge) yet gets 5.1× the median reviews (permutation $p < 0.001$); 68% are horror.
- Both tests bite: 8 of the top 10 pairs by reviews are crowded; the network test drops 3 of 35 uncrowded pairs (e.g. RPG × Visual Novel).
- **Holds up:** #1 at all 8 edge cutoffs; beats the median in 100/100 bootstraps; 3.4× without its top 5 games. **Weak spot:** #1 in 59/100.

7 Recommended action

- Week 0: pitch a first-person multiplayer horror game (#1); plan B is #2.
- Weeks 1–6: prototype. Week 7: Coming Soon page, tagged First-Person and Multiplayer first.
- Before full production: re-run on 2025 data (section 8).

8 What would change my mind

#1 and #2 are both multiplayer horror; "not crowded" is vs chance, not vs the 2023–24 co-op horror boom. If 2025 releases cross 1.5×, I switch to #3.